

Learn JavaScript: Beginner's Course

Lesson 7: Objects and JSON

Objects and JSON - Study Notes

Key Concepts

- **Object literals** – creating objects with `{}` syntax and defining properties/values.
- **Property access** – reading and writing values using dot notation (`obj.prop`) or bracket notation (`obj["prop"]`).
- **JSON (JavaScript Object Notation)** – a text based data interchange format that mirrors object literal syntax.
- **JSON.stringify** – converting a JavaScript value (object, array, primitive) to a JSON string.
- **JSON.parse** – converting a JSON string back into a JavaScript value.

Summary

In JavaScript, objects are collections of key value pairs that model real world entities. An **object literal** lets you declare an object inline using curly braces, separating each property with a comma. Properties can be accessed or modified with **dot notation** when the key is a valid identifier, or with **bracket notation** for any string key (including those with spaces or special characters).

When you need to send or store object data, you often convert it to **JSON**, a language agnostic format that looks almost identical to an object literal but is strictly text. `JSON.stringify()` takes a JavaScript value and returns its JSON representation, while `JSON.parse()` does the reverse, turning a well formed JSON string into usable JavaScript objects or primitives. Mastering these tools enables easy data exchange between client and server, configuration handling, and persistent storage (e.g., `localStorage`).

Important Points

- **Object literal syntax**: `const person = { name: "Ada", age: 30, isStudent: false };`
- **Dot vs. bracket notation**:
 - Dot: `person.name` ! `"Ada"` (only works for valid identifiers).
 - Bracket: `person["age"]` ! `30` (works for any string, also allows computed keys).
 - **Adding/updating properties**: `person.city = "London";` or `person["role"] = "Engineer";`
 - **Deleting properties**: `delete person.isStudent;`
 - **JSON restrictions**:
 - Keys must be double quoted strings.
 - Values can be strings, numbers, booleans, `null`, objects, or arrays.
 - No functions, `undefined`, `Symbol`, or trailing commas.
 - **JSON.stringify options**:
 - `JSON.stringify(value)` ! compact string.
 - `JSON.stringify(value, null, 2)` ! pretty printed with 2 space indentation.
 - Replacer function/array can filter or transform values.

- ****JSON.parse safety****: always wrap in `try/catch` when the source is untrusted to avoid syntax errors.

Examples

Example 1 – Creating and manipulating an object

```
``js
// Object literal
const car = {
  make: "Toyota",
  model: "Corolla",
  year: 2020,
  electric: false
};
// Access properties
console.log(car.make);      // Toyota
console.log(car["model"]);  // Corolla
// Add a new property
car.color = "Blue";
// Update existing property
car.year = 2021;
// Delete a property
delete car.electric;
console.log(car);
/*
{
  make: "Toyota",
  model: "Corolla",
  year: 2021,
  color: "Blue"
}
*/
``
```

Example 2 – Converting to/from JSON

```
``js
const user = {
  id: 42,
  name: "Linus",
  active: true,
  scores: [85, 92, 78],
  meta: { lastLogin: "2024-09-01T10:30:00Z" }
};
```

```
// Convert object !' JSON string (pretty printed)
const jsonStr = JSON.stringify(user, null, 2);
console.log(jsonStr);
/*
{
  "id": 42,
  "name": "Linus",
  "active": true,
  "scores": [
    85,
    92,
    78
  ],
  "meta": {
    "lastLogin": "2024-09-01T10:30:00Z"
  }
}
*/
// Convert JSON string !' object
const parsed = JSON.parse(jsonStr);
console.log(parsed.name); // Linus
console.log(Array.isArray(parsed.scores)); // true
...

```

Quick Review

1. **What are the two ways to access an object's property, and when would you prefer bracket notation?**
2. **What does `JSON.stringify` do, and what is the effect of passing `null, 2` as the second and third arguments?**
3. **Name three types of values that are *not* allowed in JSON.**
4. **How can you safely parse a JSON string that might be malformed?**
5. **Show how to add a new property `"status": "online"` to an existing object `session`.**

Further Reading

- **MDN Web Docs – Working with objects**: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Working_with_Objects
- **MDN Web Docs – JSON**: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/JSON
- **Eloquent JavaScript – Chapter 4: Data Structures: Objects and Arrays** (free online)
- **You Don't Know JS (book) – Types & Grammar**: sections on objects and JSON serialization.
- **Practice: build a small todo list app that stores its data in `localStorage` using `JSON.stringify`/`JSON.parse`.**

Happy coding! ☺=Þ€

